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Integrated circuit with conductive line having line-ends

Abstract

A semiconductor device is disclosed, including a plurality of conductive features disposed over a substrate. A dielectric layer separates the conductive features. A conductive line is provided, connecting a subset of the conductive features. The conductive line includes a line-like portion and a line-end portion.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5949111	12/1998	Morikawa	257/E21.628	H01L 21/823871
9524902	12/2015	Huang et al.	N/A	N/A
2002/0008266	12/2001	Kumagai	257/296	H10B 10/12
2002/0096723	12/2001	Awaka	257/E27.06	H01L 27/092
2006/0267096	12/2005	Maeda	N/A	N/A
2007/0181947	12/2006	Ching Ho Chan	N/A	N/A
2008/0079096	12/2007	Takahashi	N/A	N/A
2008/0217735	12/2007	Chen	257/E23.149	H01L 23/5256
2009/0011561	12/2008	Licitra	N/A	N/A
2009/0134463	12/2008	Abadeer	N/A	N/A
2009/0142701	12/2008	Hsu	N/A	N/A
2010/0095253	12/2009	Hou	N/A	N/A
2010/0283155	12/2009	Tang et al.	N/A	N/A
2011/0140203	12/2010	Hou	N/A	N/A
2011/0183518	12/2010	Usami	N/A	N/A
2011/0187004	12/2010	Park	N/A	N/A
2012/0249159	12/2011	Filippi	324/543	G01R 31/66
2013/0099296	12/2012	Lyu	N/A	N/A
2013/0270709	12/2012	Tseng et al.	N/A	N/A
2014/0191322	12/2013	Botula et al.	N/A	N/A
2015/0170959	12/2014	Huang	N/A	N/A
2017/0098574	12/2016	Ting et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2317746	12/1997	GB	H01L 21/76838

Background/Summary

(1) The present application is a continuation application of U.S. application No. 16,218,591, filed Dec. 13, 2018, which is a continuation application of U.S. application Ser. No. 15/384,076, filed Dec. 19, 2016, now U.S. Pat. No. 10,163,689, which is a divisional application of U.S. application Ser. No. 14/104,917, filed Dec. 12, 2013, now U.S. Pat. No. 9,524,902, each of which is herein incorporated by reference in its entirety.

BACKGROUND

(1) The semiconductor integrated circuit (IC) industry has experienced rapid growth. Technological advances in IC design and materials have produced generations of ICs where each generation has smaller and more complex circuits than previous generations. In the course of IC evolution, functional density (i.e., the number of interconnected devices per chip area) has generally increased while geometry size (i.e., the smallest component (or line) that can be created using a fabrication process) has decreased.

(2) This scaling down process generally provides benefits by increasing production efficiency and lowering associated costs. Such scaling down has also increased the complexity of IC processing and manufacturing. For these advances to be realized, similar developments in IC processing and manufacturing are needed. When a semiconductor device such as a metal-oxide-semiconductor field-effect transistor (MOSFET) is scaled down through various technology nodes, interconnects of conductive lines and associated dielectric materials that facilitate wiring between the transistors and other devices play a more important role in IC performance improvement. Although existing methods of fabricating IC devices have been generally adequate for their intended purposes, they have not been entirely satisfactory in all respects.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present disclosure is best understood from the following detailed description when read with the accompanying figures. It is emphasized that, in accordance with the standard practice in the industry, various features are not drawn to scale and are used for illustration purposes only. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIG. 1 is a flowchart of an example method for fabricating a semiconductor integrated circuit (IC) constructed according to various aspects of the present disclosure.

(3) FIG. 2 is a cross-sectional view of an example semiconductor device precursor according to the method of FIG. 1.

(4) FIGS. 3A and 3B are cross-sectional views of an example semiconductor integrated circuit (IC) at fabrication stages constructed according to the method of FIG. 1.

(5) FIG. 4A is a top schematic view of an example semiconductor integrated circuit (IC) at fabrication stages constructed according to the method of FIG. 1.

(6) FIGS. 4B and 4C are cross-sectional views of an example semiconductor integrated circuit (IC) at fabrication stages constructed according to the method of FIG. 1.

(7) FIGS. 4D and 4E are flowcharts of example methods for fabricating a semiconductor integrated

circuit (IC) constructed according to various aspects of the present disclosure.

(8) FIGS. 5A and 5C are top schematic views of an example semiconductor integrated circuit (IC) at fabrication stages constructed according to the method of FIG. 1.

(9) FIGS. 5B and 5D are cross-sectional views of an example semiconductor integrated circuit (IC) at fabrication stages constructed according to the method of FIG. 1.

(10) FIGS. 5E and 5F are flowcharts of example methods for fabricating a semiconductor integrated circuit (IC) constructed according to various aspects of the present disclosure.

(11) FIGS. 6A and 7A are top schematic views of an example semiconductor integrated circuit (IC) at fabrication stages constructed according to the method of FIG. 1.

(12) FIGS. 6B and 7B are cross-sectional views of an example semiconductor integrated circuit (IC) at fabrication stages constructed according to the method of FIG. 1.

(13) FIG. 6C is a flowchart of an example method for fabricating a semiconductor integrated circuit (IC) constructed according to various aspects of the present disclosure.

DETAILED DESCRIPTION

(14) The following disclosure provides many different embodiments, or examples, for implementing different features of the invention. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(15) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, elements described as being “below” or “beneath” other elements or features would then be oriented “above” the other elements or features. Thus, the exemplary term “below” can encompass both an orientation of above and below. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(16) FIG. 1 is a flowchart of one embodiment of a method **100** of fabricating one or more semiconductor devices according to aspects of the present disclosure. The method **100** is discussed in detail below, with reference to a semiconductor device **300** shown in FIGS. 2, 3, 4A-4B, 5A-5C, 6 and 7A-7B for the sake of example. It is understood that additional steps can be provided before, during, and after the method, and some of the steps described can be replaced or eliminated for other embodiments of the method.

(17) Referring to FIGS. 1 and 2, the method **100** begins at step **102** by receiving a semiconductor device precursor **200**. The semiconductor device precursor **200** includes a substrate **210**. In the present embodiment, the substrate **210** includes silicon. In alternative embodiments, the substrate may include germanium, silicon germanium, gallium arsenide or other appropriate semiconductor materials. Alternatively and for some embodiments, the substrate **210** may include an epitaxial layer. For example, the substrate **210** may have an epitaxial layer overlying a bulk semiconductor. Further, the substrate **210** may be strained for performance enhancement. For example, the epitaxial layer may include a semiconductor material different from those of the bulk semiconductor such as a layer of silicon germanium overlying bulk silicon or a layer of silicon

overlying a bulk silicon germanium formed by a process including selective epitaxial growth (SEG). Furthermore, the substrate **210** may include a semiconductor-on-insulator (SOI) structure such as a buried dielectric layer. Also alternatively, the substrate **210** may include a buried dielectric layer such as a buried oxide (BOX) layer, such as that formed by a method referred to as separation by implantation of oxygen (SIMOX) technology, wafer bonding, SEG, or other appropriate methods. In fact various embodiments may include any of a variety of substrate structures and materials.

(18) The semiconductor device precursor **200** may also include various isolation features **220**. The isolation features **220** separate various device regions in the substrate **210**. The isolation features **220** include different structures formed by using different processing technologies. For example, the isolation features **220** may include shallow trench isolation (STI) features. The formation of a STI may include etching a trench in the substrate **210** and filling in the trench with insulator materials such as silicon oxide, silicon nitride, or silicon oxynitride. The filled trench may have a multi-layer structure such as a thermal oxide liner layer with silicon nitride filling the trench. A chemical mechanical polishing (CMP) may be performed to polish back excessive insulator materials and planarize the top surface of the isolation features **220**.

(19) The semiconductor device precursor **200** also includes one or more first conductive features **230**. In one embodiment, the first conductive feature **230** may include high-k/metal gates (HK/MGs), a three-dimension HK/MGs wrapping over a fin-like structure. As an example, the HK/MGs may include a gate dielectric layer and metal gate (MG). The gate dielectric layer may include LaO, AlO, ZrO, TiO, Ta.sub.2O.sub.5, Y.sub.2O.sub.3, SrTiO.sub.3 (STO), BaTiO.sub.3 (BTO), BaZrO, HfZrO, HfLaO, HfSiO, LaSiO, AlSiO, HfTaO, HfTiO, (Ba,Sr)TiO.sub.3 (BST), Al.sub.2O.sub.3, Si.sub.3N.sub.4, oxynitrides (SiON), or other suitable materials. The MG may include a single layer or multi layers, such as a metal layer, a liner layer, a wetting layer, and an adhesion layer. The MG may include Ti, Ag, Al, TiAlN, TaC, TaCN, TaSiN, Mn, Zr, TiN, TaN, Ru, Mo, Al, WN, Cu, W, or any suitable materials. Additionally, sidewall spacers (not shown) are formed on the sidewalls of the HK/MGs.

(20) In another embodiment, the first conductive features **230** include electrodes, capacitors, resistors or a portion of a resistor. In yet another embodiment, the first conductive features **230** include a portion of the interconnect structure. For example, the first conductive features **230** include contacts, metal vias, or metal lines.

(21) The semiconductor device precursor **200** also includes second conductive features **240** in the substrate **210**. A top surface to the second conductive feature **250** may not be at a same horizontal level as a top surface of the first conductive feature **230**. In one embodiment, the second conductive features **240** include doped regions (such as sources or drains). In another embodiment, the second conductive features **240** include electrodes, capacitors, resistors or a portion of a resistor, or a portion of the interconnect structure.

(22) The semiconductor device precursor **200** also includes a first dielectric layer **250** deposited over the substrate **210**, including between/over each of the first conductive features **230** and over the second conductive features **240**. The first dielectric layer **250** includes silicon oxide, silicon nitride, oxynitride, a dielectric material having a dielectric constant (k) lower than thermal silicon oxide (therefore referred to as low-k dielectric material layer), or other suitable dielectric material layer. The first dielectric layer **250** includes a single layer or multiple layers. A CMP may be performed to remove excessive the first dielectric layer **250** to expose the top surface of the first conductive features **230**, as well as to provide a substantially planar top surface for the first conductive features **230** and the first dielectric layer **250**.

(23) The semiconductor device precursor **200** also includes a second dielectric layer **260** over the first conductive features **230** and the first dielectric layer **250**. The second dielectric layer **410** is similar in many respects to the first dielectric layer **250** discussed above.

(24) The semiconductor device precursor **200** also includes conductive plugs **270** in the first and

second dielectric layers, **250** and **260**, to form full contacts extending down to the first and the second conductive features, **230** and **240**. The conductive plugs **270** are formed along a first direction. The conductive plugs **270** may include copper (Cu), aluminum (Al), tungsten (W), copper or copper alloy, such as copper magnesium (CuMn), copper aluminum (CuAl) or copper silicon (CuSi), or other suitable conductive material. In some cases, the conductive plug **270** is referred to as a metal plug.

(25) Referring to FIGS. **1** and **3**, once the semiconductor device precursor **200** is received, the method **100** proceeds to step **104** by depositing a third dielectric layer **310** over the second dielectric layer **260**, including over the conductive plugs **270**, and a hard mask (HM) layer **320** over the third dielectric layer **310**. The third dielectric layer **310** is similar in many respects to the first dielectric layer **250** discussed above in association with FIG. **2**.

(26) The HM layer **320** includes silicon nitride, silicon oxide, silicon carbide, titanium oxide, titanium nitride, tantalum oxide, tantalum nitride, or any suitable materials. In addition, the HM layer **320** may include a single layer or multiple layers. The HM layer **320** is different from the third dielectric layer **310** to achieve etching selectivity during a subsequent etch, which will be described later. The HM layer **320** may be deposited by suitable techniques, such as CVD, or physical vapor deposition (PVD). In one embodiment, the HM layer **320** includes a stack of layers: a bottom titanium nitride layer deposited over the third dielectric layer **310**; an anti-reflection coating (ARC) layer deposited over the bottom titanium nitride layer and a top titanium nitride layer deposited over the ARC layer, as shown in FIG. **3B**.

(27) Referring to FIGS. **1** and **4A-4B**, the method **100** proceeds to step **106** by forming a first opening **410** on the HM layer **320**. The first opening **410** has a first length $L_{sub.1}$ landing in a plane, which is perpendicular to the first direction, and a first width $W_{sub.1}$. In the present embodiment, the first length $L_{sub.1}$ is defined by two line-ends **412** and is substantially larger than the first width $W_{sub.1}$. Therefore, in some cases, the first opening **410** is referred to as a line-like opening. The first opening **410** is aligned to a subset of the conductive plugs **270**, now labeling them with the reference number **270A**. The first opening **410** may be formed by a procedure (shown in FIG. **4D**) that includes depositing a resist layer (block **450**), forming a first resist pattern on the resist layer, and etching the HM layer **320** using the first resist pattern as an etch mask (block **452**). The first resist pattern is then removed by wet stripping or plasma ashing (block **454**). In one embodiment, the first openings **410** are formed by removing the top titanium nitride layer of the HM stack **320** using the first resist pattern as an etch mask, as shown in FIG. **4C** and FIG. **4E** (blocks **460**, **462**, and **464**).

(28) Referring to FIGS. **1** and **5A-5C**, the method **100** proceeds to step **108** by forming a second opening **510** on the HM layer **320**, landing in the same place as (or near to) where the first opening **410** landed. In the present embodiment, the second opening **510** fully contacts, or overlaps with, the first opening **410** at the line-ends **412**. Thus, the second opening **510** forms a new line-end **512** for the first opening **410**. In other words, with the second opening **510**, the line-end **412** is replaced by the new line-ends **512**. The second opening **510** has a second length $L_{sub.2}$ along the same direction as the first length $L_{sub.1}$ and a second width $W_{sub.2}$ along the same direction as the first width $W_{sub.1}$. The second length $L_{sub.2}$ is substantially smaller than the first length $L_{sub.1}$. The second width $W_{sub.2}$ is same or larger than the first width $W_{sub.1}$. In some cases, the second opening **510** is referred to as a line-end opening.

(29) In one embodiment, the second width $W_{sub.2}$ is larger than the first width $W_{sub.1}$ and the second opening **510** locates symmetrically to the first opening **410** at the line-end **412** (as shown in FIG. **5A**). In another embodiment, the second width $W_{sub.2}$ is larger than the first width $W_{sub.1}$ and the second opening **510** locates asymmetrically to the first opening **410** at the line-end **412** (as shown in FIG. **5C**). In some embodiments, the second opening **510** has a rectangular shape.

(30) As shown in FIG. **5E**, the second openings **510** may be formed similar in many respects to the first opening **410** discussed above in association with FIGS. **4A** and **4B**, which includes forming a

resist layer over the HM layer **320** and the first opening **410**, forming a second resist pattern on the resist layer (block **550**) and etching the HM layer **320** using the second resist pattern as an etch mask (block **552**). The second resist pattern is then removed by wet stripping or plasma ashing (block **554**). In one embodiment, the second openings **510** are formed by removing the top titanium nitride layer of the HM stack **320** using the second resist pattern as an etch mask, as shown in FIGS. 5D and 5F (blocks **560**, **562**, and **564**).

(31) Referring to FIGS. **1** and **6A-6B**, the method **100** proceeds to step **110** by forming a dielectric trench **610** to expose the conductive plugs **270A**. In the present embodiment, the dielectric trench **610** is formed by etching the third dielectric layer **310** using the HM layer **320** with the first and the second openings, **410** and **510**, as an etch mask. In one embodiment, the etch process selectively removes the third dielectric layer **310** but substantially does not etch the HM layer **320**. Therefore the dielectric trench **610** carries a profile of a combination of the first and the second openings, **410** and **510**. In another embodiment (shown in FIG. **6C**), using the HM layer **320** with the first and the second openings, **410** and **510**, in the top titanium nitride layer of the HM stack **320** as an etch mask, the dielectric trench **610** is formed by etching the ARC layer (block **650**), the bottom titanium nitride layer (block **652**) and the third dielectric layer **310** (block **654**).

(32) The dielectric trench **610** has a line-like portion **612** and a line-end portion **614**. The line-end portion **614** has the second width $W_{sub.2}$, which is equal or larger than the first width $W_{sub.1}$ of the line portion **612**. In one embodiment, the second width $W_{sub.2}$ is larger than the first width $W_{sub.1}$ and the line-end portion **614** located symmetrically to the line-like portion **612**. In another embodiment, the second width $W_{sub.2}$ is larger than the first width $W_{sub.1}$ and the line-end portion **614** locates asymmetrically to the line-like portion **612**.

(33) Referring to FIGS. **1** and **7A-7B**, the method **100** proceeds to step **112** by filling in the dielectric trench with conductive materials and forming a conductive line **710**. In one embodiment, a barrier layer is filled in the dielectric trench **610** first by a proper deposition technique, such as PVD and CVD. The barrier layer may include a metal and is electrically conductive but does not permit (or reduces) inter-diffusion and reactions between the third dielectric material layer **310** and the conductive material to be filled in the dielectric trench **610**. The barrier layer may include refractory metals and their nitrides. In various examples, the barrier layer includes TiN, TaN, Co, WN, TiSiN, TaSiN, or combinations thereof. The barrier layer may include multiple films.

(34) The conductive material **710** then fills in the dielectric trench **610**, as well as over the barrier layer. The conductive material **710** may include copper (Cu), aluminum (Al), tungsten (W), copper or copper alloy, such as copper magnesium (CuMn), copper aluminum (CuAl) or copper silicon (CuSi), or other suitable conductive material. The conductive material **710** may be deposited by y PVD, CVD, metal-organic chemical vapor deposition (MOCVD), or plating. A recess is then performed to etch back the excessive conductive material **710**, as well as the excessive barrier layer, to form the conductive line **710** and a substantial planar surface with the third dielectric layer **310**. For example, the recess is performed by chemical mechanical polishing (CMP).

(35) Referring again to FIG. **7A**, in the present embodiment, the conductive line **710** carries the profile of the dielectric trench **610** and has a line portion **712** and a line-end portion **714**. The line-end portion **714** has the second width $W_{sub.2}$, which is equal or larger than the first width $W_{sub.1}$ of the line portion **712**. In one embodiment, the second width $W_{sub.2}$ is larger than the first width $W_{sub.1}$ and the line-end portion **714** located symmetrically to the line portion **712**. In another embodiment, the second width $W_{sub.2}$ is larger than the first width $W_{sub.1}$ and the line-end portion **714** locates asymmetrically to the line portion **712**.

(36) Additional steps can be provided before, during, and after the method **100**, and some of the steps described can be replaced, eliminated, or moved around for additional embodiments of the method **100**. The device **300** may undergo further CMOS or MOS technology processing to form various features and regions.

(37) Based on the above, the present disclosure offers a semiconductor device. The semiconductor

device may include a plurality of conductive features over a substrate, a dielectric layer separating the conductive features, and a conductive line connecting a subset of the conductive features. The conductive line may include a line-like portion having a first length and a first width and a line-end portion connecting to an end of the line-like portion, wherein the line-end portion has a second width, equal to or larger than the first width, wherein the line-end portion has a second length, smaller than the first length.

(38) In some embodiments, the line-end portion of the conductive line stays symmetrically to the line-like portion. The line-end portion of the conductive line, having a larger width than the width of the line-like portion, may stay asymmetrically to the line-like portion. The line-end portion may have a rectangular shape. The plurality of conductive features may extend through the dielectric layer. The semiconductor device may include a hard mask (HM) layer disposed over the dielectric layer, wherein the conductive line is formed within the dielectric layer and the HM layer.

(39) In other embodiment, a semiconductor device is provided, which may include a substrate, a first set of conductive features formed over the substrate, a second set of conductive features formed over the substrate, a first dielectric layer separating the first set of conductive features and the second set of conductive features, a second dielectric layer formed over the first dielectric layer, and a hard mask (HM) layer formed over the second dielectric layer, wherein the HM layer and the second dielectric layer comprise a line-end opening and a line-like opening, wherein the line-end opening is aligned to the first set of conductive features, wherein the line-end opening connects with the line-like opening at a first end of the line-like opening, wherein the line-like opening has a first width and the line-end opening has a second width greater than the first width.

(40) In some embodiments, a conductive line is provided connecting the first set of conductive features and the second set of conductive features. The conductive line may be disposed in a trench within the second dielectric layer. The line-like opening may be aligned to the second set of conductive features. The line-end opening may have a rectangular shape. A second line-end opening may be provided, connected with the line-like opening at a second end of the line-like opening. In some embodiments, the HM layer includes a stack of layers, including a bottom titanium nitride layer, an anti-reflection coating (ARC) layer formed over the bottom titanium nitride layer, and a top titanium nitride layer formed over the ARC layer.

(41) In another embodiment, a semiconductor device is provided, including a substrate, a first dielectric layer formed over the substrate, a second dielectric layer formed over the first dielectric layer, a set of conductive features extending through the first and second dielectric layers, a third dielectric layer formed over the second dielectric layer, and a hard mask (HM) layer formed over the third dielectric layer, wherein a trench is formed within the third dielectric layer and the HM layer, the trench comprising a line-like opening and two line-end openings connecting with the line-like opening at respective ends of the line-like opening, wherein the two line-end openings are aligned to a first subset of the set of conductive features, wherein the line-like opening has a first width and the two line-end openings each have a second width greater than the first width.

(42) In some embodiments, the trench is filled with conductive material to form a conductive line. The conductive line may connect at least two conductive features within the set of conductive features. The line-like opening may be aligned to a second subset of the set of conductive features. A length and a width of each line-end opening may be the same. The line-like opening may have a first length and the line-end openings each may have a second length smaller than the first length. In some embodiments, the HM layer includes a stack of layers, including a bottom titanium nitride layer, an anti-reflection coating (ARC) layer formed over the bottom titanium nitride layer, and a top titanium nitride layer formed over the ARC layer.

(43) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages

of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A device comprising: a first dielectric layer disposed over a substrate; a first conductive feature disposed within the first dielectric layer; a second conductive feature disposed within the first dielectric layer; a third conductive feature over the substrate, wherein the first conductive feature is over and electrically coupled to the third conductive feature; a fourth conductive feature in the substrate, wherein the second conductive feature extends continuously and physically contacts the fourth conductive feature, wherein the third conductive feature and the fourth conductive feature comprise a gate electrode and a doped region, respectively, and wherein the third conductive feature and the fourth conductive feature are arranged along a first direction; and a conductive line connected to the first conductive feature and the second conductive feature, the conductive line including: a line-like portion having a first length along the first direction and a first width along a second direction, wherein the line-like portion is aligned to the first conductive feature and the second conductive feature, and wherein the line-like portion is directly above, vertically overlaps, and physically contacts the second conductive feature, and a first line-end portion connected to the line-like portion and disposed over the first conductive feature, wherein the first line-end portion has a second width along the second direction that is larger than the first width, wherein the first line-end portion has a second length along the first direction that is smaller than the first length.
2. The device of claim 1, wherein the conductive line further includes a second line-end portion connecting to the line-like portion, wherein the second line-end portion has a third width that is larger than the first width, wherein the second line-end portion has a third length that is smaller than the first length.
3. The device of claim 2, wherein the third width is the same as the second width.
4. The device of claim 2, wherein the first line-end portion is connected to a first end of the line-like portion, and wherein the device further comprises a fifth conductive feature directly under and in direct contact with the second line-end portion.
5. The device of claim 2, wherein the first line-end portion physically contacts the first conductive feature.
6. The device of claim 1, wherein at least one of the first conductive feature and the second conductive feature is a contact.
7. The device of claim 1, wherein the fourth conductive feature comprises only one doped region.
8. A device comprising: a first dielectric layer disposed over a substrate; a doped feature contact and a gate contact extending through the first dielectric layer; a gate stack disposed over the substrate, wherein the gate contact extends to the gate stack; a doped feature disposed in the substrate and adjacent the gate stack, wherein the doped feature contact extends to the doped feature; and a conductive line in contact with the doped feature contact and the gate contact, wherein no other conductive lines are between the conductive line and the gate contact and the doped feature contact, the conductive line including: a line-like portion having a first length and a first width and in physical contact with the doped feature contact, a first line-end portion connected to a first end of the line-like portion, wherein the first line-end portion has a second width that is larger than the first width, wherein the first line-end portion has a second length that is smaller than the first length, and a second line-end portion connected to a second end of the line-like portion, wherein the second line-end portion has a third width that is larger than the first width, wherein the second line-end portion has a third length that is smaller than the first length.
9. The device of claim 8, wherein the second line-end portion is disposed over the gate contact.

10. The device of claim 8, wherein the line-like portion is aligned to and overlaps the doped feature contact and the gate contact from a top view.
 11. The device of claim 8, wherein the line-like portion physically contacts a top surface of the doped feature contact, the top surface of the doped feature contact facing away from the substrate, and wherein the second line-end portion physically contacts a top surface of the gate contact, the top surface of the gate contact facing away from the substrate.
 12. The device of claim 8, wherein the conductive line is formed of a metal containing material and a barrier layer material at least partially surrounding the metal containing material.
 13. The device of claim 8, wherein the second width is equal to the third width, and wherein the second length is equal to the third length.
 14. A device comprising: a substrate; a gate stack disposed over the substrate; a doped feature disposed in the substrate, wherein the gate stack and the doped feature are arranged along a first direction from a top view; a first dielectric layer disposed over the substrate; a first conductive feature and a second conductive feature extending through the first dielectric layer, wherein the first conductive feature extends to the gate stack and the second conductive feature extends to the doped feature; and a conductive line disposed over the substrate and in physical contact with the first conductive feature and the second conductive feature, the conductive line comprising: a line-like portion having a first length along the first direction and a first width along a second direction that is different than the first direction, a first line-end portion having a second width along the second direction that is larger than the first width, and a second length along the first direction that is smaller than the first length, and a second line-end portion having a third width along the second direction that is larger than the first width, and a third length along the first direction that is smaller than the first length, wherein a first end of the line-like portion is connected to the first line-end portion and a second end of the line-like portion opposite the first end is connected to the second line-end portion.
 15. The device of claim 14, wherein the second line-end portion is disposed over the second conductive feature.
 16. The device of claim 14, wherein the first line-end portion physically contacts a top surface of the first conductive feature, the top surface of the first conductive feature facing away from the substrate, and wherein the second line-end portion physically contacts a top surface of the second conductive feature, the top surface of the second conductive feature facing away from the substrate.
 17. The device of claim 14, wherein the conductive line is formed of a metal containing material and a barrier layer material at least partially surrounding the metal containing material.
 18. The device of claim 14, wherein the line-like portion physically contacts the second conductive feature.
 19. The device of claim 14, wherein the second conductive feature extends to only one continuous doped feature.
 20. The device of claim 8, wherein the gate stack and the doped feature are arranged along a direction from a top view, and wherein the line-like portion is oriented lengthwise along the direction.
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